IIA

LCC - FCEIA

Laboratorio

Deep Learning Desde Cero

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Objetivos

Reconocimiento de Texto Manuscrito

↪ queremos leer documentos manuscritos

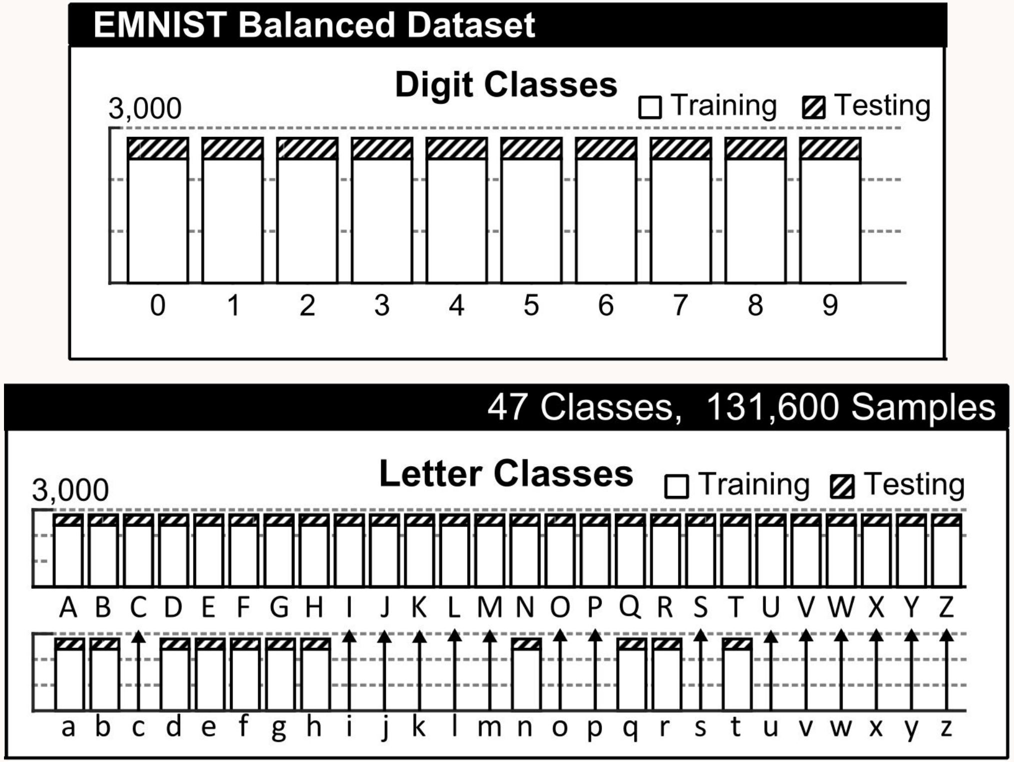
↪ tenemos imágenes de letras y números

↪ necesitamos reconocer letras y números

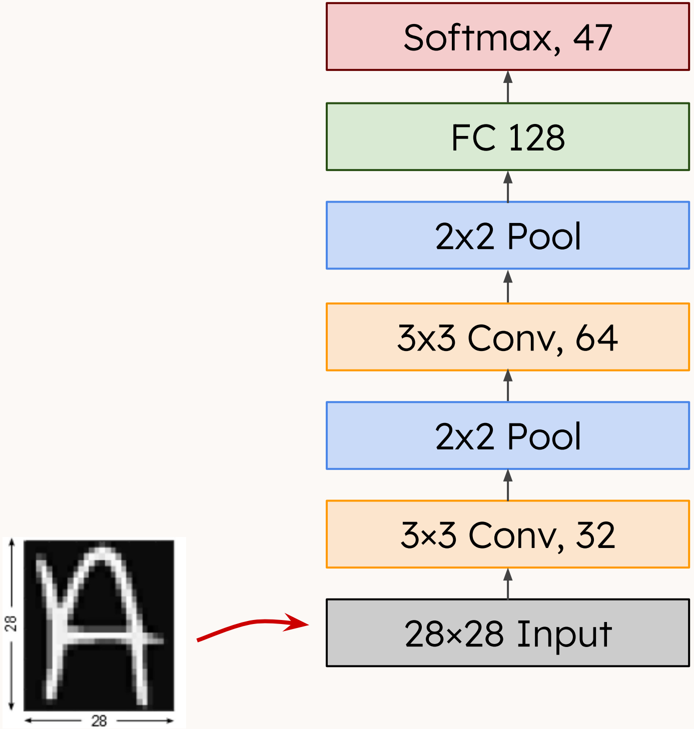
↪ clasificación de imágenes

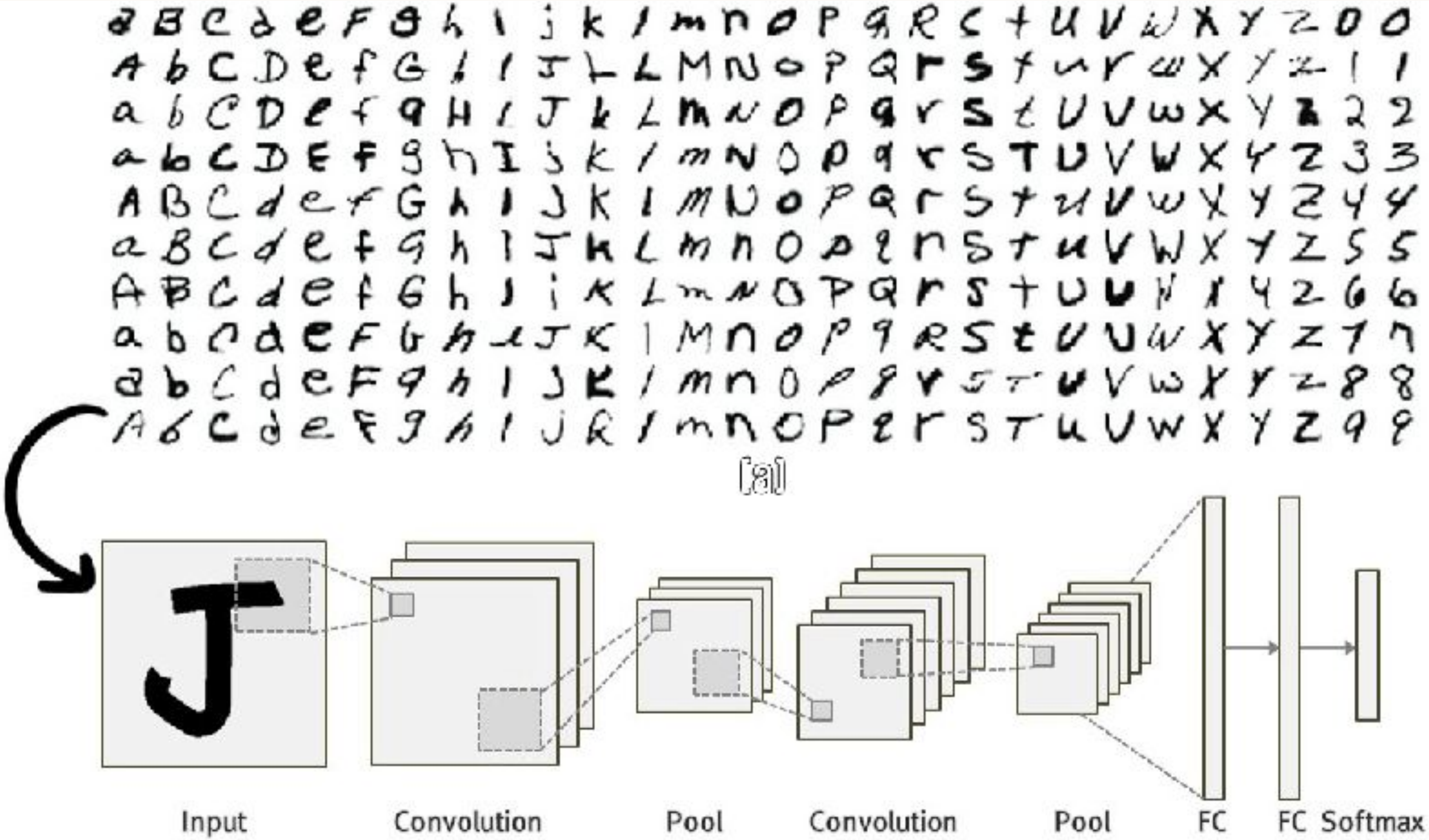
↪ cada letra y cada número será una clase

Dataset


EMNIST Balanced Dataset
Digit Classes
3,000
Training
Testing
0
1
2
3
4
5
6
7
8
9
47 Classes, 131,600 Samples
Letter Classes
3,000
Training
Testing
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
a b c d e f g h i j k l m n o p q r s t u v w x y z

Modelo


Softmax, 47
FC 128
2x2 Pool
3x3 Conv, 64
2x2 Pool
3x3 Conv, 32
28×28 Input
28
28



(a)

(b)

Métricas y Optimizadores

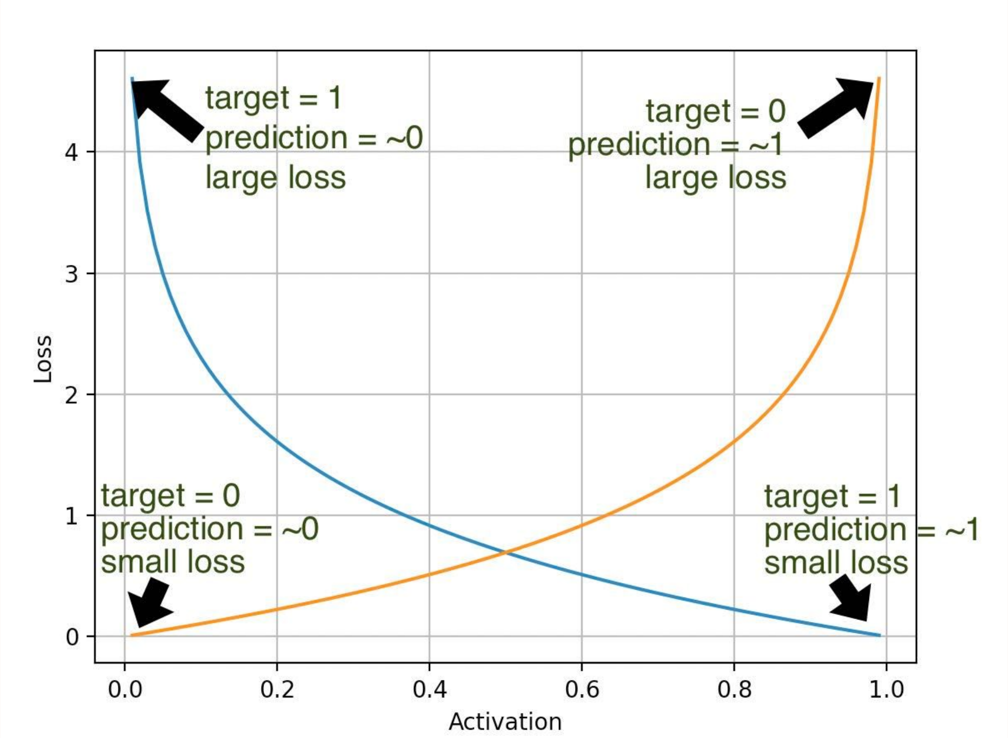
$$\text{Loss} = -\frac{1}{\substack{\text{output}\\ \text{size}}} \sum_{i=1}^{\substack{\text{output}\\ \text{size}}} y_i \cdot \log \hat{y}_i + (1 - y_i) \cdot \log (1 - \hat{y}_i)$$

prob. of class i

$$\text{Loss} = -\sum_{i=1}^{\substack{\text{output}\\ \text{size}}} y_i \cdot \log \hat{y}_i$$

Métricas y Optimizadores

log(y)
&
log(1-y)



Métricas y Optimizadores

Gradient Descent (SGD)

weights

$$w_t = w_{t-1} - \alpha \cdot dw_t$$

learning rate

Métricas y Optimizadores

SGD w/Momentum

moving average

$$v_t = \beta v_{t-1} + (1 - \beta) dw_t$$

$$w_t = w_{t-1} - \alpha v_t$$

momentum rate

Métricas y Optimizadores

AdaGrad

element-wise squares

$$v_t = v_{t-1} + dw_t^2$$

$$w_t = w_{t-1} - \frac{\alpha}{\sqrt{v_t + \varepsilon}} dw_t$$

momentum depends on square root

Métricas y Optimizadores

RMSProp

exponential weights

$$v_t = \beta v_{t-1} + (1 - \beta) dw_t^2$$

$$w_t = w_{t-1} - \frac{\alpha}{\sqrt{v_t} + \varepsilon} dw_t$$

Métricas y Optimizadores

Adam

$$v_t = \beta_1 v_{t-1} + (1 - \beta_1) dw_t$$

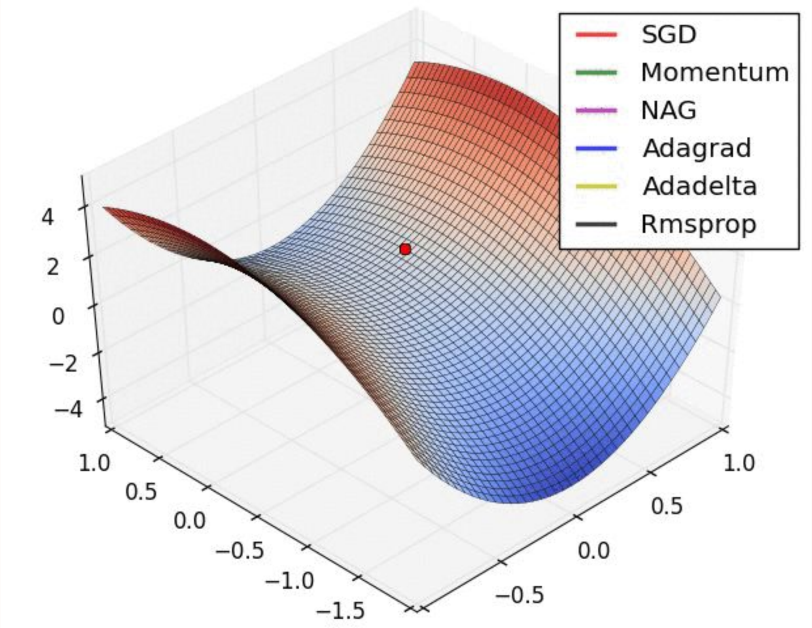
$$s_t = \beta_2 s_{t-1} + (1 - \beta_2) dw_t^2$$

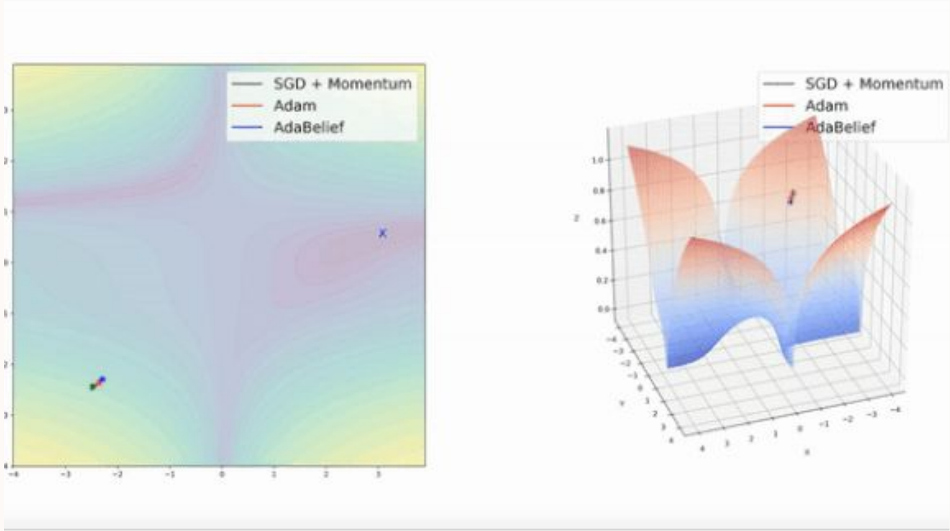
momentum

$$w_t = w_{t-1} - \frac{\alpha \hat{v}_t}{\sqrt{\hat{s}_t} + \varepsilon} dw_t$$

RMSProp

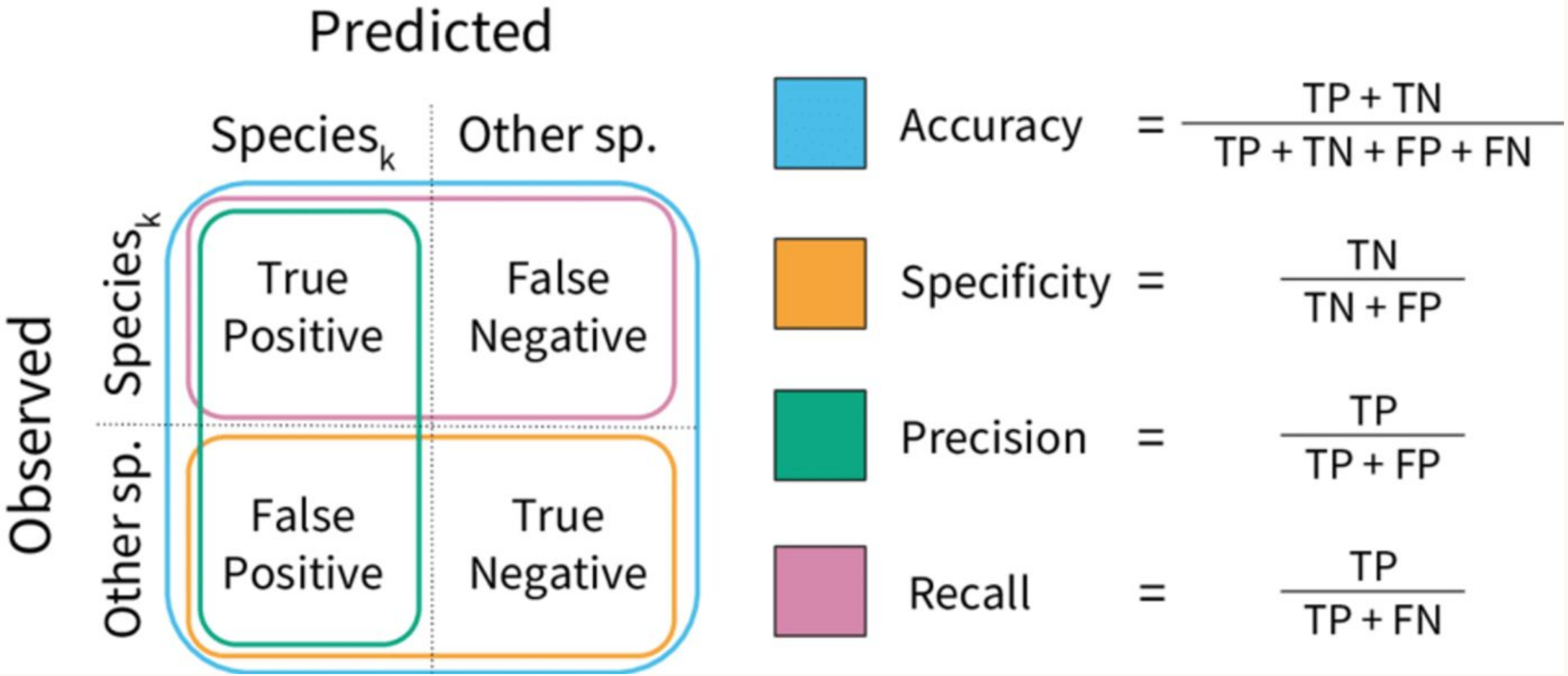
Métricas y Optimizadores





Entrenamiento

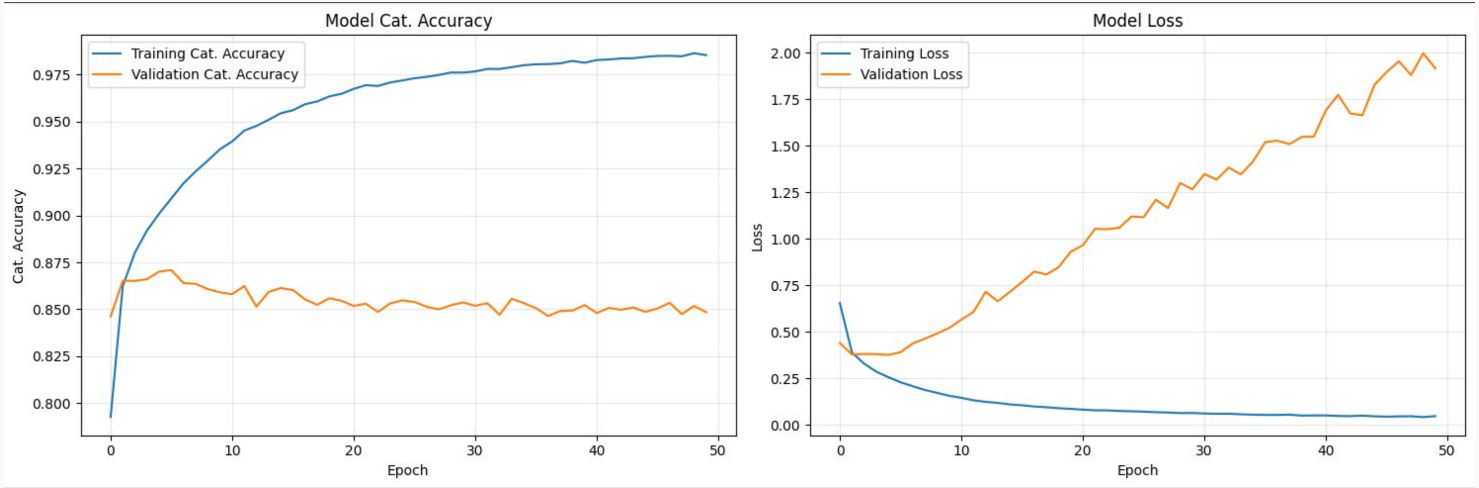
Evaluación

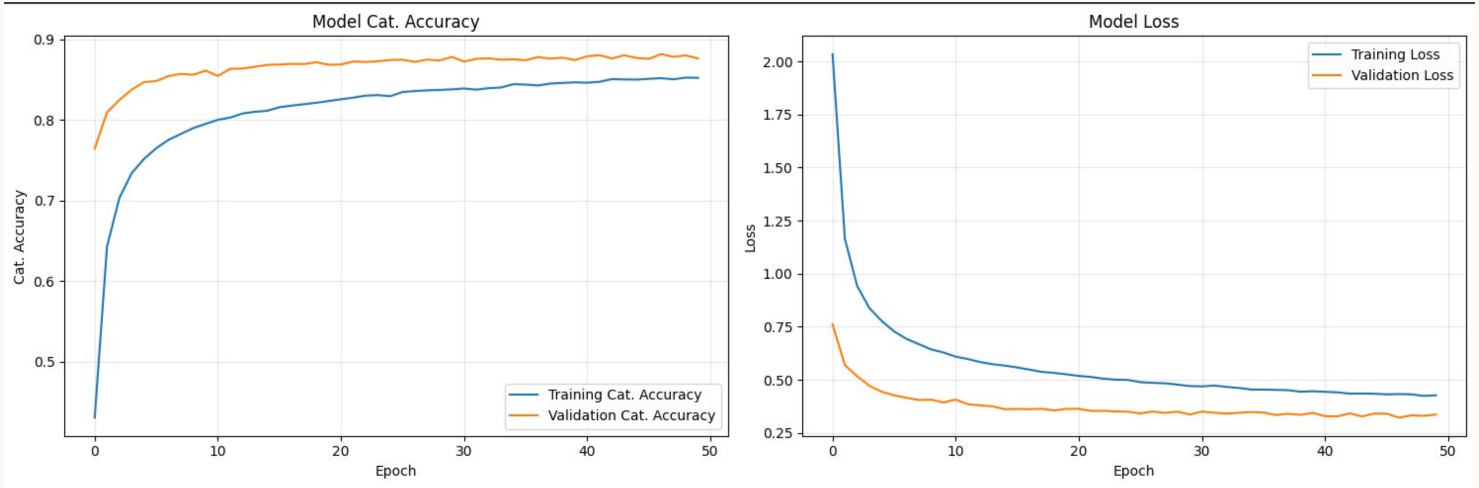


Predicted
Species$_k$
Other sp.
Observed
Species$_k$
Other sp.
True Positive
False Negative
False Positive
True Negative
Accuracy = $\frac{TP + TN}{TP + TN + FP + FN}$
Specificity = $\frac{TN}{TN + FP}$
Precision = $\frac{TP}{TP + FP}$
Recall = $\frac{TP}{TP + FN}$

¿Resultados?

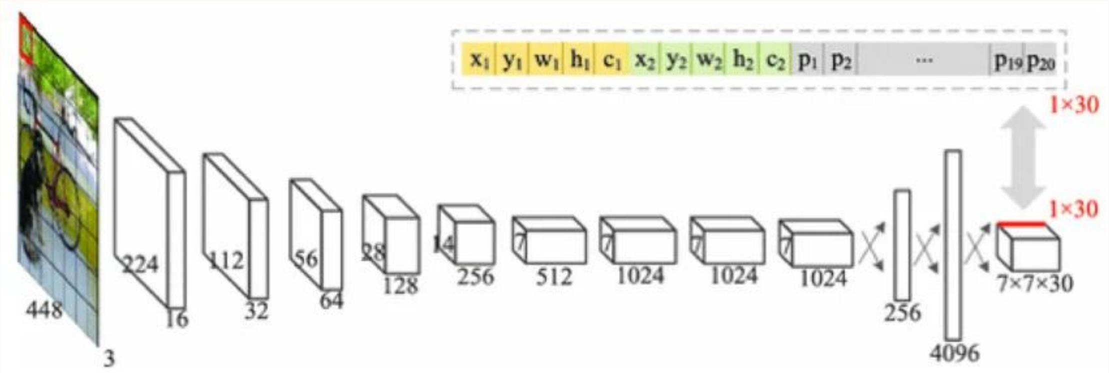
Naïve



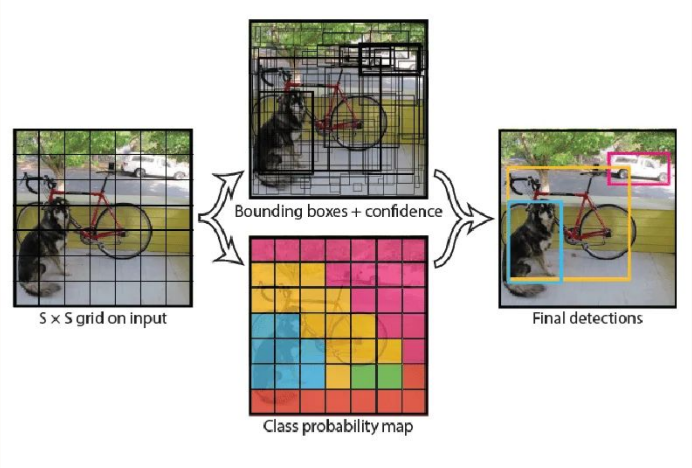


Regularized

YOLOv1



x_1 y_1 w_1 h_1 c_1 x_2 y_2 w_2 h_2 c_2 p_1 p_2 ... p_{19} p_{20}
1×30
1×30
448
3
224
16
112
32
56
64
28
128
14
256
7
512
1024
1024
1024
256
4096
7×7×30



S × S grid on input
Bounding boxes + confidence
Class probability map
Final detections